

# Week 10

## Acoustics

## Sound Propagation and Noise Barriers

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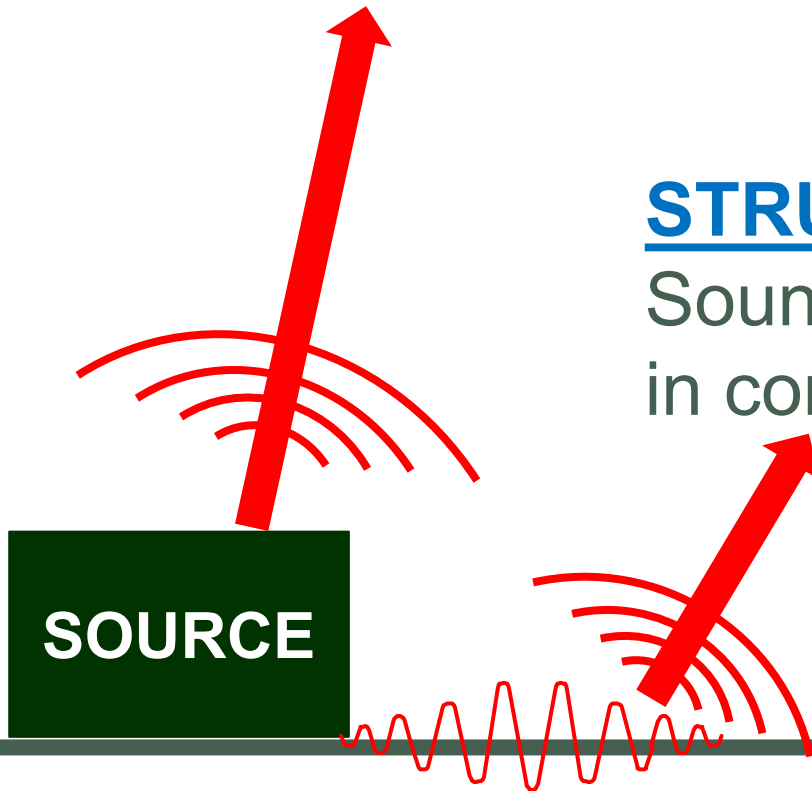
# SOUND PROPAGATION

## AIRBORNE SOUND

Sound emitted directly from source and propagates through air

## STRUCTURE-BORNE SOUND

Sound emitted from surfaces in contact with vibrating source



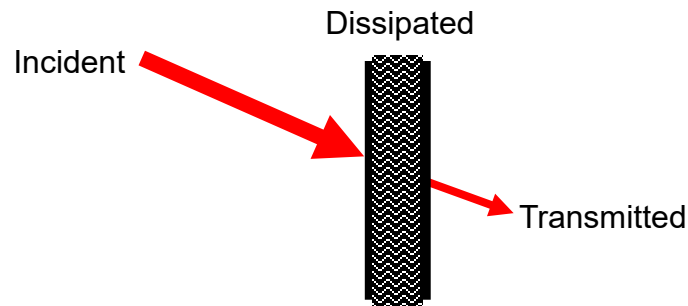
# SOUND PROPAGATION

- **Airborne Sound:** a sound that arrives at a listener location by propagation *through the air*.
  - Examples: speech, musical instruments, air sound from ducts, sound from building equipment within the space
- **Structure-borne Sound:** originates in an *impact* or *vibration producing source* that is in contact with a building component which works as an transmitter and amplifier of the source. The impact or vibrating source itself may not create much sound.
  - Examples: sound from building equipment heard several floors away, water pipes touching structure, footfalls on a hard surface

# BASIC NOISE CONTROL APPROACHES

## Noise (Airborne)

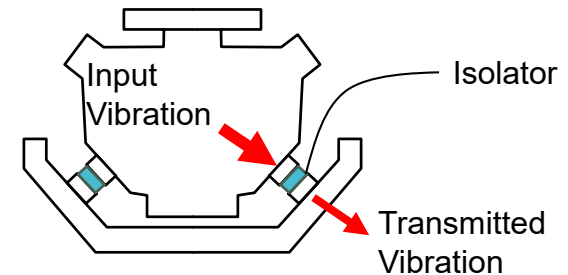
### Barriers



Block Transmission of Noise

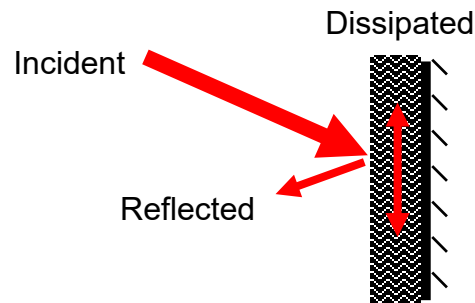
## Vibration (Structure-borne)

### Isolation



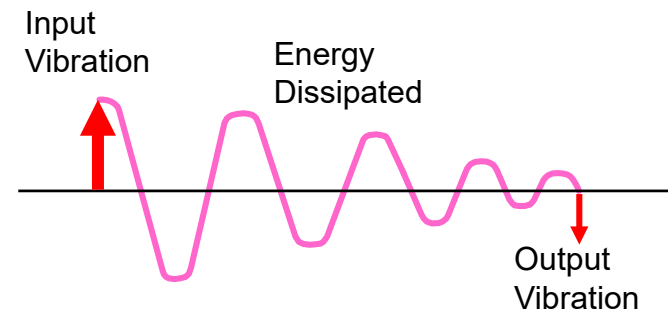
Block Transmission of Vibration

### Absorbers



Absorb/Minimize Reflected Noise

### Damping



Absorb Vibrational Energy

# AIRBORNE NOISE CONTROL

## BARRIER versus ABSORBER

### ISOLATION/BARRIER:

Primary function is *to block airborne noise*. It should be dense, impermeable and have mass. Examples are sheet metal, glass, drywall, wood, concrete, etc.

### ABSORBER:

Primary function is *to absorb airborne noise* that would otherwise reflect off surfaces. It should be porous, and is generally lightweight. Examples are foam, fiberglass wall panels, drapery, carpeting, etc.

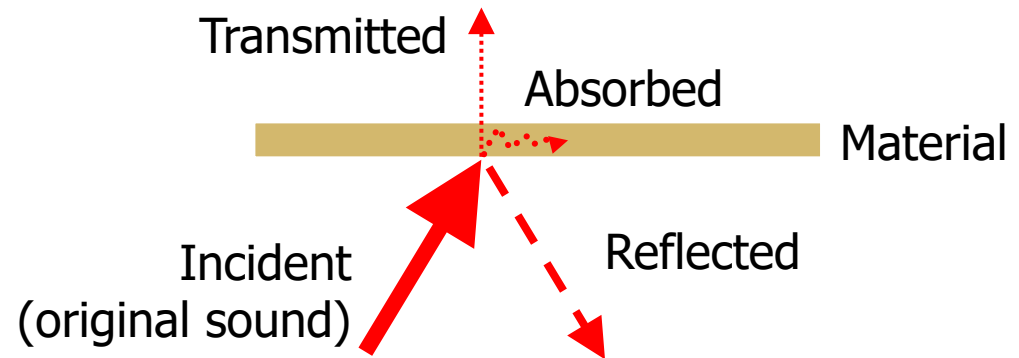
GOOD BARRIERS ARE GENERALLY BAD ABSORBERS  
AND VICE-VERSA

Airborne  
Sound Isolation  
(*BLOCKING SOUND*):  
NR & TL

# BLOCKING SOUND

- **Blocking sound using barriers** is most often the first consideration for noise control – used for defining spaces and rooms.
- Noise source and receiver are **separated** by:
  - Walls and partitions
  - Floors and ceilings
  - Doors, windows and access panels
  - Enclosures
  - Outdoor barriers (e.g., highway barrier or berm)
- Careful consideration must be made when **choosing the materials** for construction of the barriers.
- Just as equally important are the **design details** of the barrier (joints, openings, attachments, ventilation, etc.)

# BOUNDARY PHENOMENA



- Incident sound on a boundary (such as a room surface) can be:
  - *Reflected* (either an echo or diffused sound)
  - *Transmitted*
  - *Absorbed* as heat within the material
    - Very small – consider that a 95 dB sound power wave (loud!) contains only 0.0032 watts  
( $95 = 10 \log(0.0032/10^{-12})$ )
    - Neglect heat effect in calculations
- Most often, all three occur

# SOUND TRANSMISSION LOSS

- When a sound wave hits the surface of a building component, part of the wave is absorbed ( $\alpha$ ), part is reflected ( $\rho$ ) & part is transmitted ( $\tau$ ).

$$\alpha + \rho + \tau = 1$$

where:

$\alpha$  (Alpha) = absorption coefficient

$\rho$  (Rho) = reflection coefficient

$\tau$  (Tau) = transmission coefficient

- Common values of  $\tau$  :

|                  |                                   |
|------------------|-----------------------------------|
| walls & floors   | $10^{-2}$ to $10^{-8}$ (unitless) |
| ¼" glass sheet   | $\approx 7.8 \times 10^{-4}$      |
| 6" concrete slab | $6.3 \times 10^{-6}$              |

Due to the small numbers of the transmission coefficient, it is more common to use Transmission Loss (TL).

# SOUND TRANSMISSION LOSS

**Transmission Loss (TL):** “a measure of how much sound energy is reduced in transmission through materials” in units of dB. Egan, p. 175

$$TL = 10 \log (1/\tau)$$

where:  $\tau$  = transmission coefficient

*The larger the TL value, the greater the acoustic isolation capability of the material/partition.*

TL is a function of frequency

# SOUND TRANSMISSION LOSS

In theory, the Transmission Loss can vary between 0 and infinity. In practice, its maximum is usually much lower ( $\approx 65\text{-}85$  dB at high frequencies, less at low frequencies) due to sound leaks and physical limitations of the barrier.

$$TL = 10 \log (1/\tau)$$

Therefore, when  $\tau$  is “1” (**all sound transmitted**), as through an **open window**,

$$TL = \log (1/\tau) = \log (1) = 0$$

When  $\tau$  approaches “0”, (**no transmission**), as for a very thick, **heavy building component**

$$TL = \log (1/\tau) = \log (1/0) \rightarrow \infty$$

# SOUND TRANSMISSION LOSS

**The Partition Noise Reduction (NR)** can be calculated by taking the difference between the dB levels on either side of a panel.

$$NR = L_{p1} - L_{p2}$$

where

NR is the noise reduction (dB)

$L_{p1}$  is the sound level in the source room (dB)

$L_{p2}$  is the sound level in the receiver room (dB)

Note that this Noise Reduction (NR) is across a partition. Also, Noise Reduction Coefficient (NRC) is a sound absorption term, and they are *unrelated*.

# SOUND TRANSMISSION LOSS

**Transmission Loss** is calculated from the following equation.

$$TL = NR - 10 \log (A_2/S)$$
$$TL = (L_{p1} - L_{p2}) - 10 \log (A_2/S)$$

where

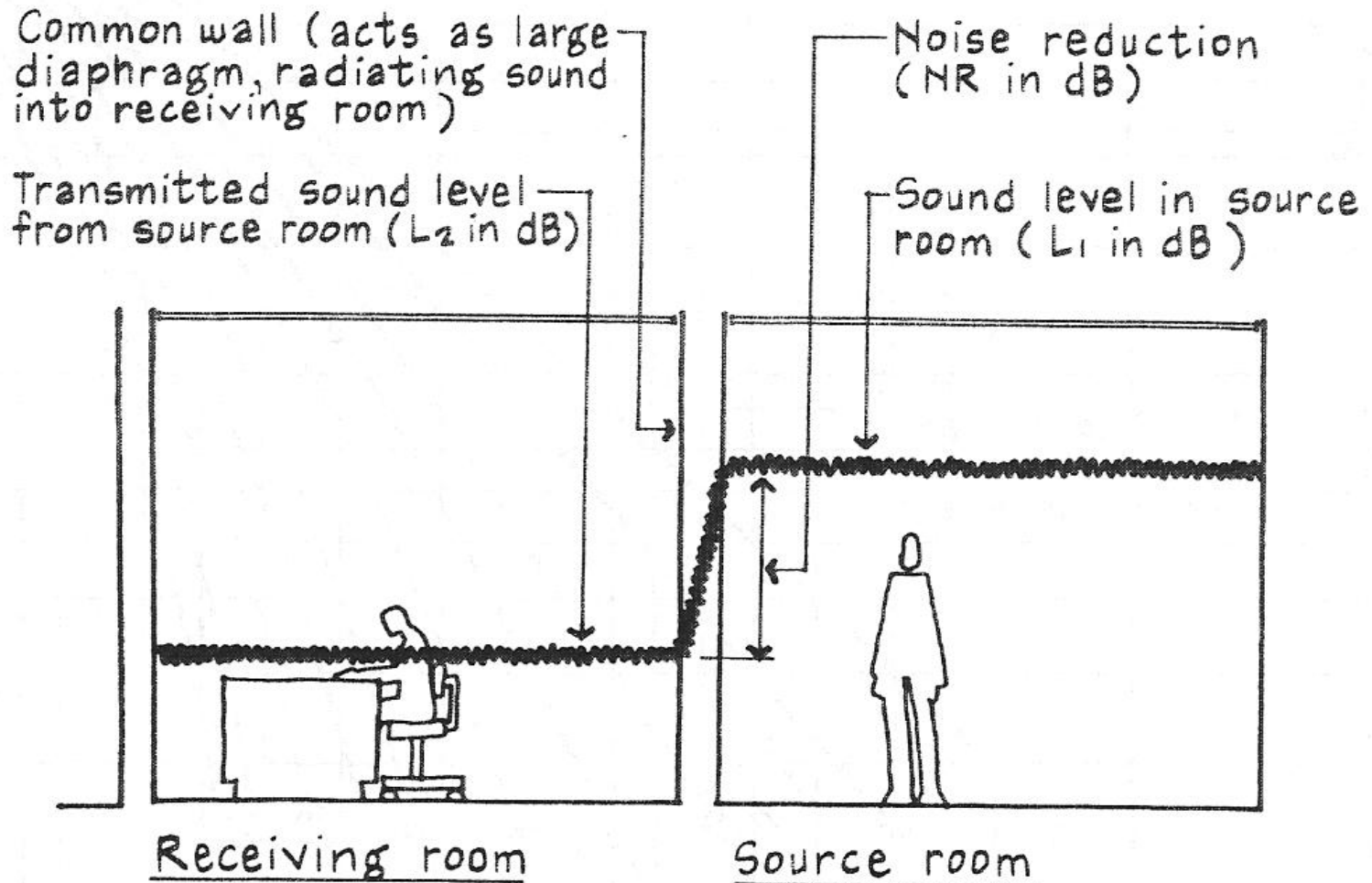
**A<sub>2</sub>** is the sound absorption in the receiving room (Sabin)  
**S** is the surface area of the common barrier wall (sq. ft.)

Dependent on:

- area of demising wall
- sound absorption in receiving room
- difference in sound pressure level in both spaces

# SOUND TRANSMISSION LOSS

## Transmission Loss



# SOUND TRANSMISSION LOSS

## What is the difference between NR and TL?

- Remember that a SPL depends on the environment in which it was measured.
  - A highly reflective room would produce a different SPL than a highly absorptive receiving room. This affects the NR of the barrier wall.

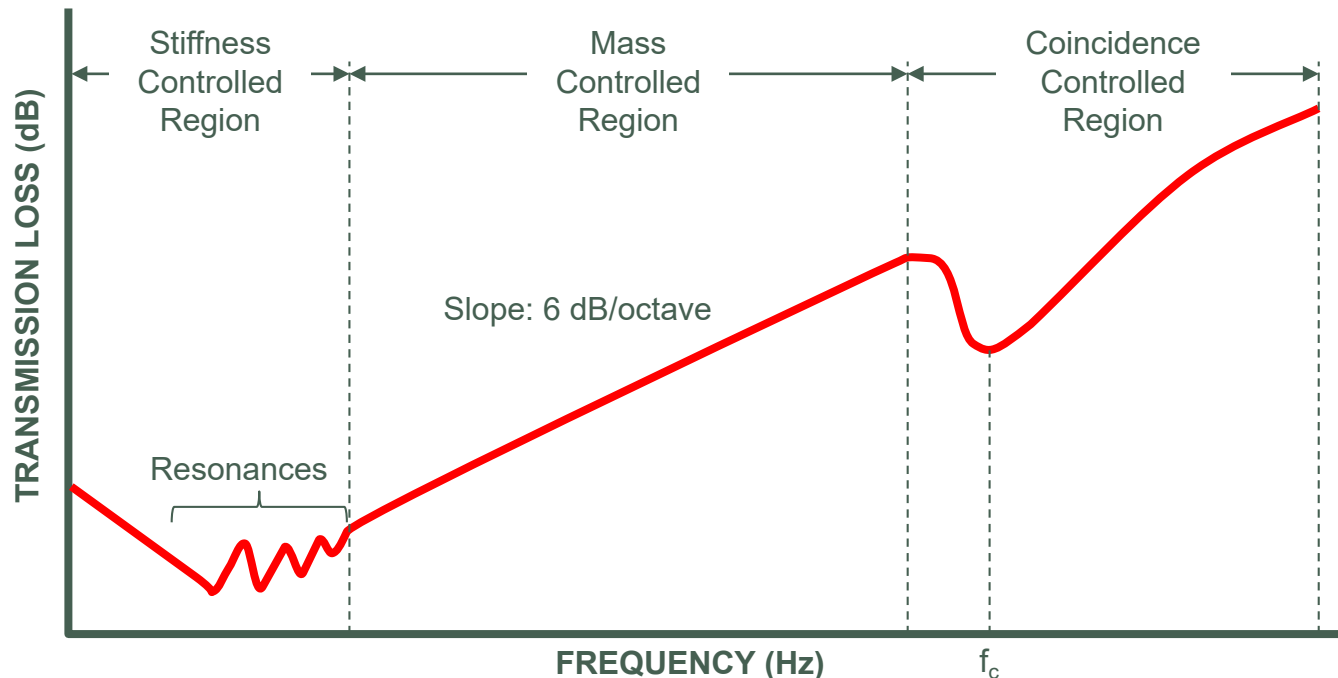
$$NR = L_{p1} - L_{p2}$$

- Transmission Loss is an inherent property of the construction, and thus the room effect must be normalized out of the measurement through adjusting for the receiving room characteristics.

$$TL = NR - 10 \log(A_2/S)$$

# TL OF SINGLE-LEAF PANELS

- **Single-leaf panels**
  - a.k.a. single-wall or single-layer barrier
- Examples include:
  - One layer of drywall, plywood, sheet metal, glass, concrete or paneling
- There are **three frequency regions of response**
  - stiffness, mass and coincidence (damping) controlled



# TL OF SINGLE-LEAF PANELS

Single-leaf panels obey the **Mass Law** and **Coincidence Effect**

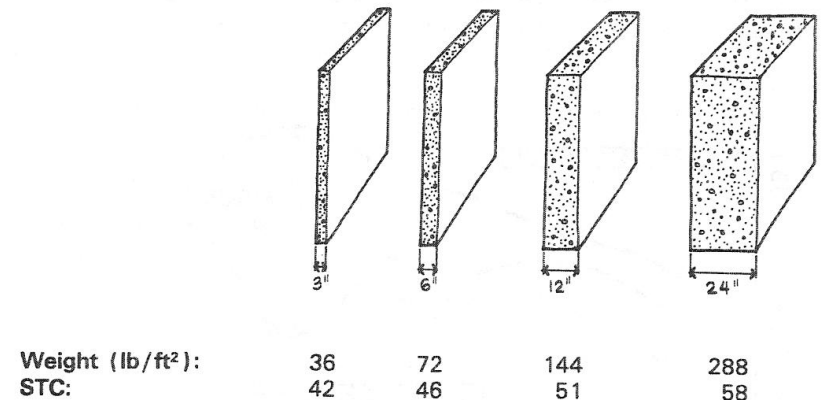
- **Mass Law:** *doubling the surface mass* or *doubling the frequency* gives a *theoretical 6 dB* increase in the TL of a panel

$$TL = 20 \log(f) + 20 \log(W) - C$$

$f$  = frequency (Hz)

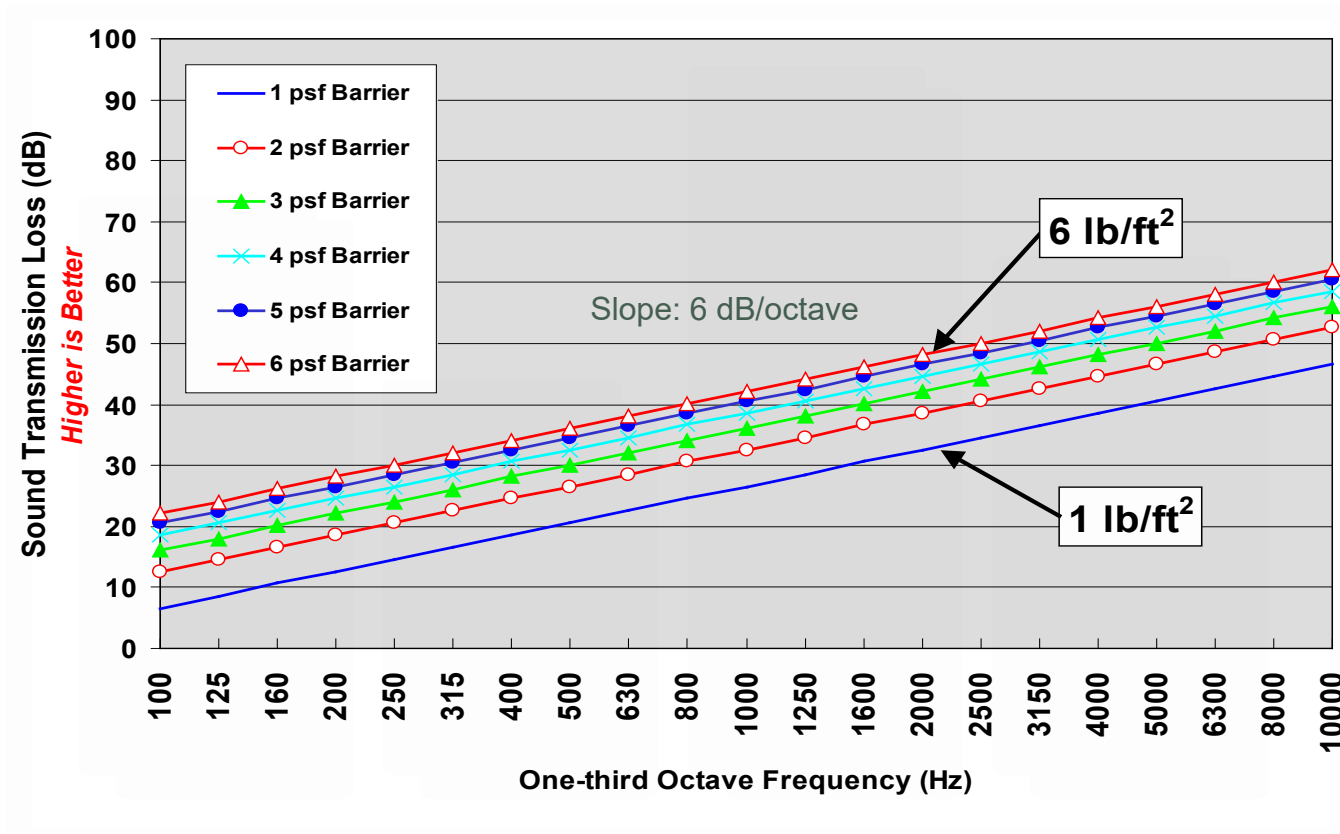
$W$  = surface density (lb/ft<sup>2</sup> or kg/m<sup>2</sup>)  
(a.k.a. surface weight)

$C = 33$  if  $W$  is lb/ft<sup>2</sup>,  
 $47$  if  $W$  is kg/m<sup>2</sup>



# TL OF SINGLE-LEAF PANELS

## MASS-CONTROLLED REGION:



Mass-Law:  $STL = 20 \cdot \log(\text{surface weight}) + 20 \cdot \log(\text{frequency}) - 33$   
Surface weight is in units of lb/ft<sup>2</sup> (psf)

STL increases by 6 dB as surface weight doubles.  
This quickly reaches a point of diminishing returns.

# TL OF SINGLE-LEAF PANELS

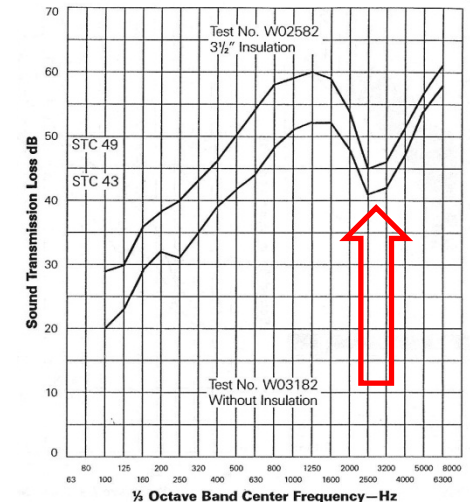
- In practice, the increase is approximately 5 dB
- Surface mass and stiffness of a panel are related for most materials
  - An increase in mass (giving a higher TL) increases the stiffness (giving a lower TL)
  - This opposing effect on TL results in a real world deviation from the mass law
- Further mass law deviation results from damping
  - Damping is the energy-absorbing property of a panel
  - Damping materials dissipate sound and vibration through the working of their elastic medium

# TL OF SINGLE-LEAF PANELS

- The panel TL also is a strong function of the natural frequency of the panel which causes resonance, in this case: **coincidence dip**
  - **Natural frequency** – the frequency that an object tends to vibrate at when hit, plucked or otherwise disturbed, and is a characteristic of that object
  - **Resonance** – the reinforcement of a particular frequency when the source of sound or vibration is equal to the natural frequency of an object or space

# TL OF SINGLE-LEAF PANELS

- When the resonant bending frequency of the single-leaf panel is encountered and sound of that frequency is incident upon the panel, there is a reduction in the TL called the *coincidence dip* or *coincidence effect*
- The coincidence dip can be 15 dB  
→ Very significant
- The coincidence frequency is a function of panel thickness, density and modulus of elasticity. It increases in frequency when the panel thickness decreases.
  - While decreasing panel thickness may be attractive to move the coincidence dip higher, the reduced TL at low frequencies (due to less mass) does not typically justify it

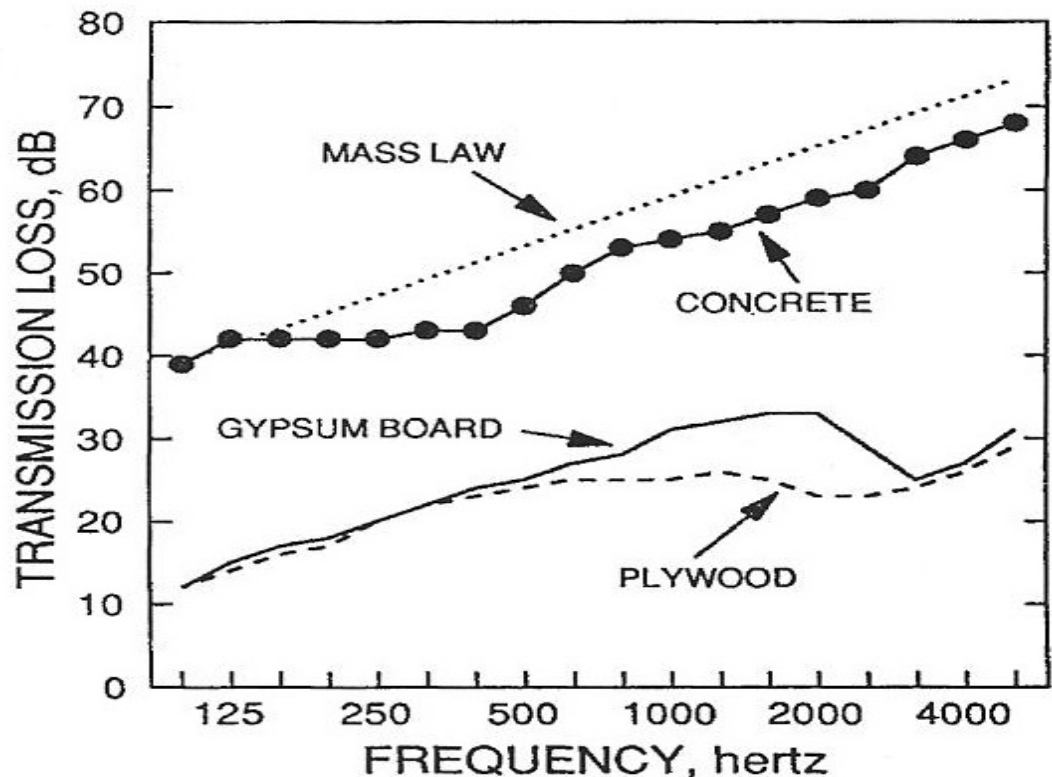


# TL OF SINGLE-LEAF PANELS

Since most building materials are stiff and relatively thick, the coincidence effect can be expected

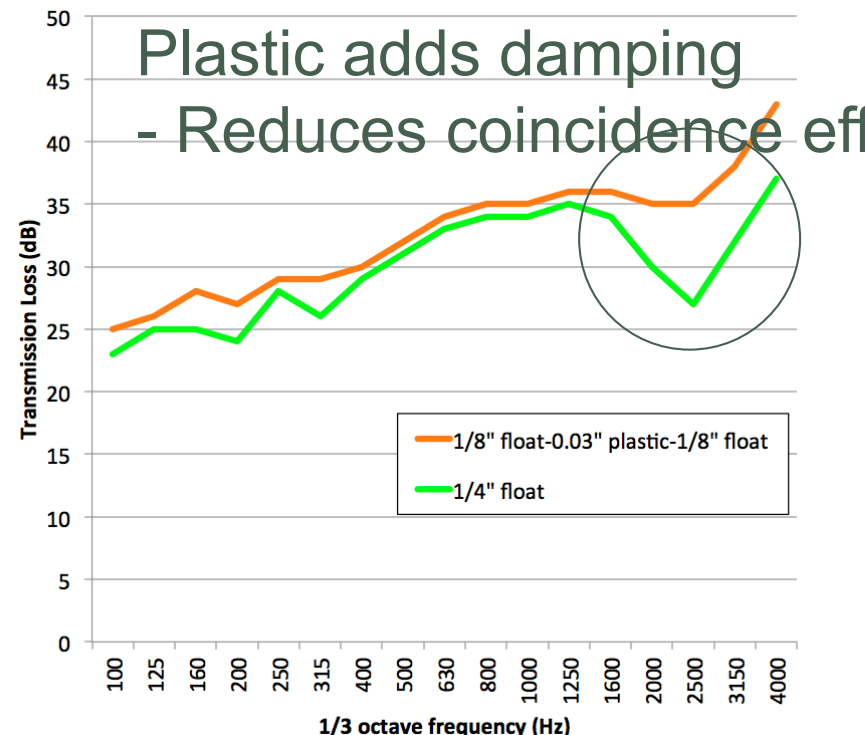
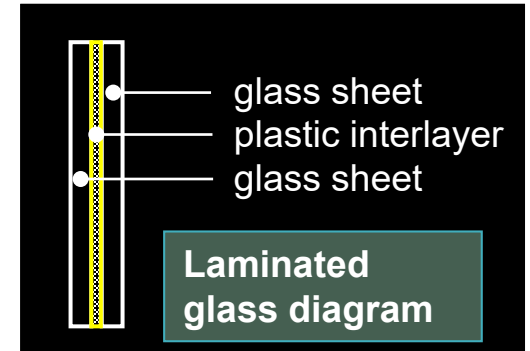
These occur at different frequencies for different materials

- Dip: concrete @ 200 Hz, 5/8" ply @ 1600 Hz, 1/2" gyp @ 3150 Hz, 1/8" glass (not shown) @ 5000 Hz



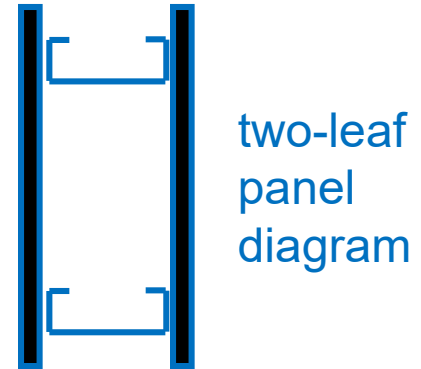
# TL OF SINGLE-LEAF PANELS

- Increasing the damping of the panel flattens the coincidence dip effect.
- Damped panels have a higher TL than an untreated panel of the same thickness.
  - Internal damping: inherent in the material, such as steel (low damping) or lead (high damping).
  - External / Added damping: viscoelastic materials applied to a surface or sandwiched between panels (more effective of the two).
- Damping has the most effect in the coincidence controlled region. This is where the coincidence dip occurs.



# TL OF TWO-LEAF PANELS

- Two-leaf panels (a.k.a. double-walls) are composed of two separate, *ideally uncoupled*, leaves.
- In the real world, each leaf of a two-leaf panel are coupled by:
  - Studs
  - Floor & ceiling
  - Air (acts as a spring)
  - Sway bracing
- A two-leaf panel generally has a higher TL than a single-leaf panel of the same weight
  - Sometimes a narrow cavity depth in a two-leaf panel will produce worse TL than a single-leaf panel



# TL OF TWO-LEAF PANELS

## Mass-air-mass resonance

- Air trapped in the cavity acts like a spring, transfers vibrational energy from one side to the other
- Causes a increase in TL around the resonant frequency
- Decrease the resonant frequency by:
  - Increase the mass of the layer(s)
  - Increase the separation between layers (i.e., deeper cavity)
  - Add a porous absorber to the cavity

$$f_{m-a-m} = K \sqrt{\frac{m_1 + m_2}{dm_1 m_2}}$$

$f_{m-a-m}$  = mass-air-mass resonance frequency, Hz

$m_1$  = surface mass of first layer (kg/m<sup>2</sup>)

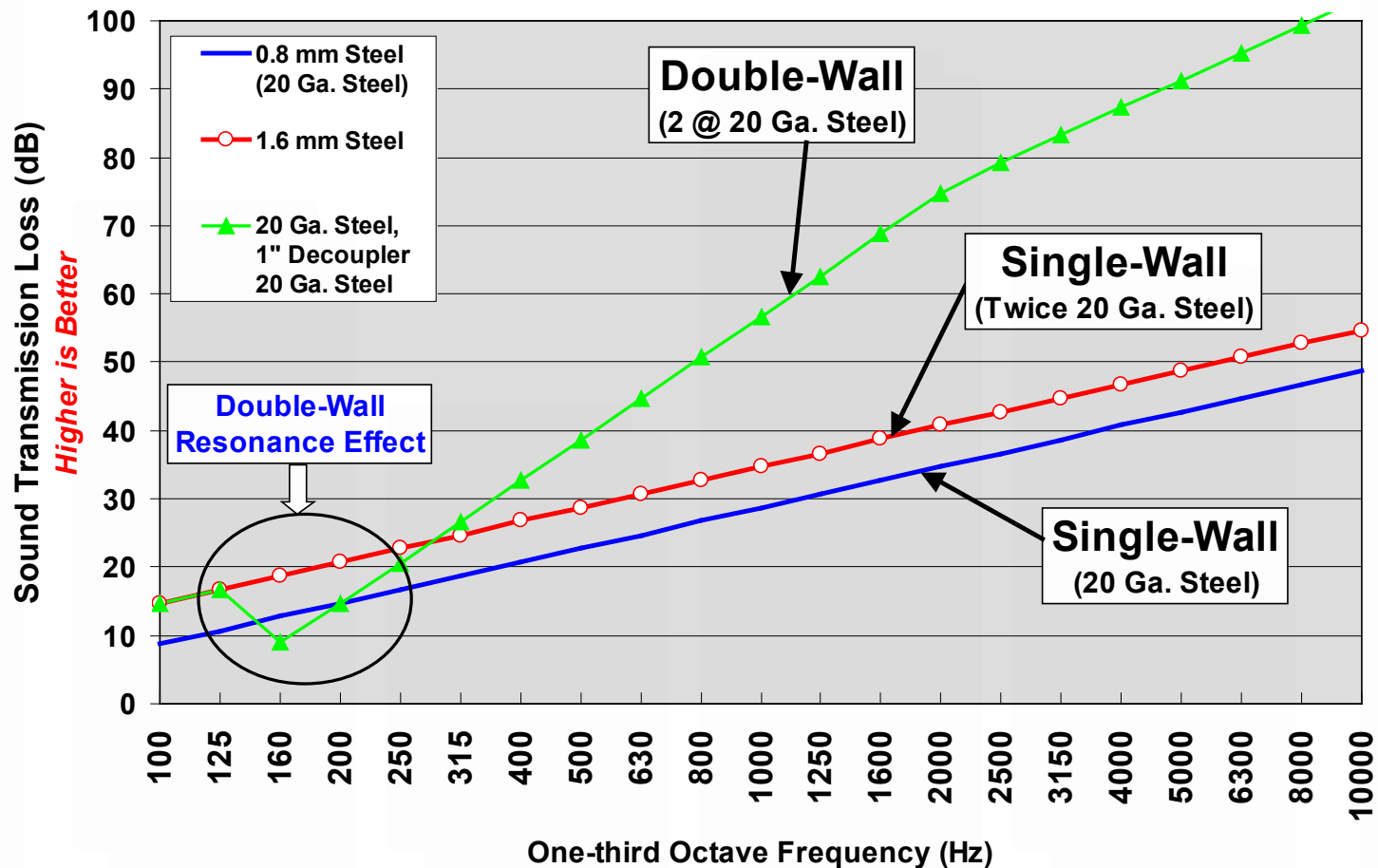
$m_2$  = surface mass of second layer (kg/m<sup>2</sup>)

$d$  = separation between layers (mm)

$K$  = 60 for an empty cavity,

43 for a cavity with sound absorber

# TL OF TWO-LEAF PANELS



Instead of doubling the thickness of a single-wall barrier, it is substantially better to separate two single-walls and create a double-wall barrier system. Note, however, that there is a degradation in performance around the double-wall resonance.

# TL OF TWO-LEAF PANELS

How can improvements be made to a two-leaf panel?

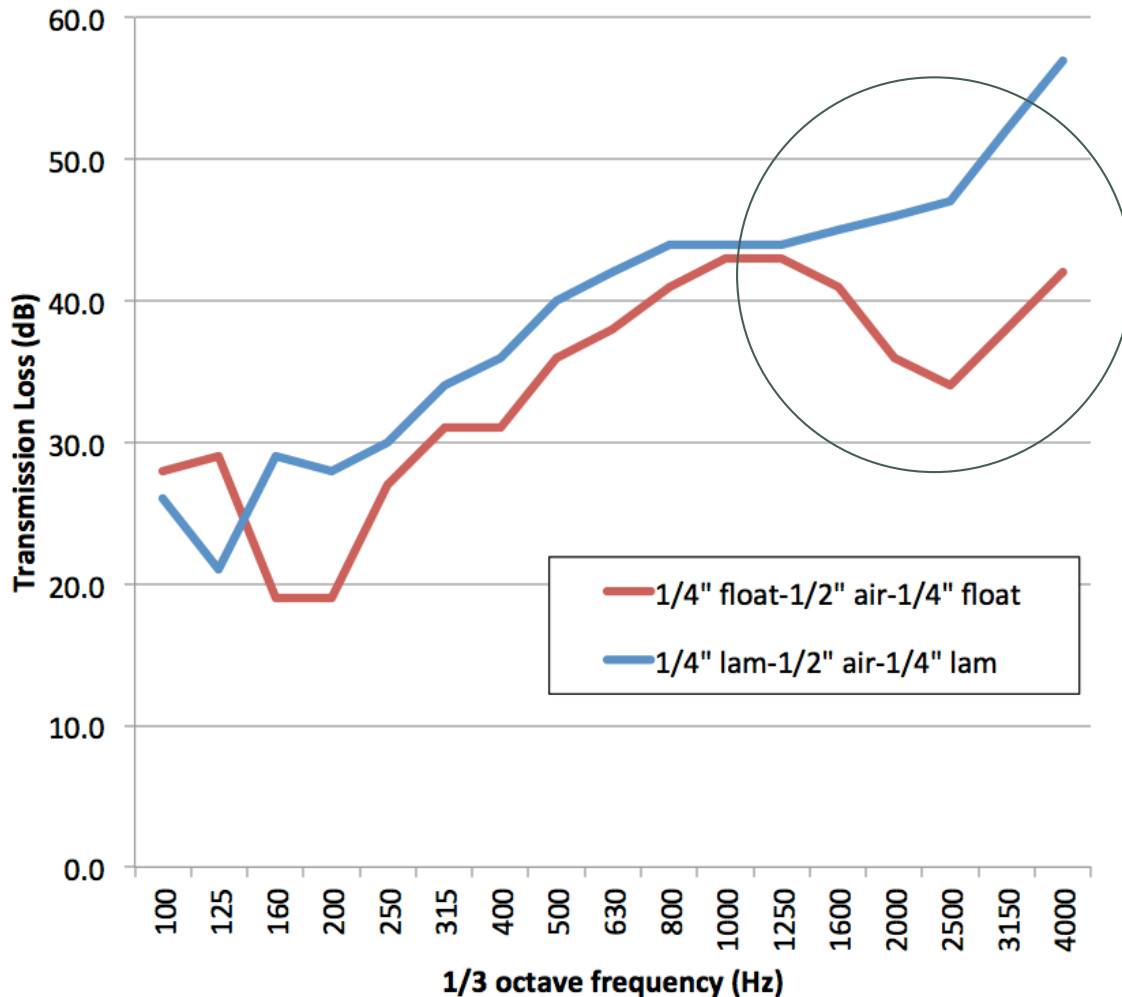
- *Decoupling (vibration isolating) the leaves* through resilient mounting, staggered studs or double studs
- *Adding mass* through multiple layers of gypsum board
- *Dissimilar leaves* will help to decrease the effect of the coincidence dip because the panel thicknesses are different (typically accomplished by use of multiple layers of gypsum board)
  - e.g., 2 layers gypsum on one side of studs and 3 layers on other side

# TL OF TWO-LEAF PANELS

- How can improvements be made to a two-leaf panel? (cont.)
  - Adding a *porous absorber* in the cavity helps to reduce the additional dip caused by the resonant frequency of the cavity
    - Filling cavities with a thickness up to  $\frac{3}{4}$  of the cavity thickness is recommended with  $\frac{1}{2}$  of the cavity a minimum.
  - *Increasing the cavity depth* will decrease the cavity resonance, possibly out of the frequency region of interest, by decreasing the air stiffness in the cavity
  - Using a *damped material* decreases coincidence dip
  - *Decreasing the stiffness* through use of lightweight gauge steel studs or steel over wood

# TL OF TWO-LEAF PANELS

- Damping helps with two-leaf panels also



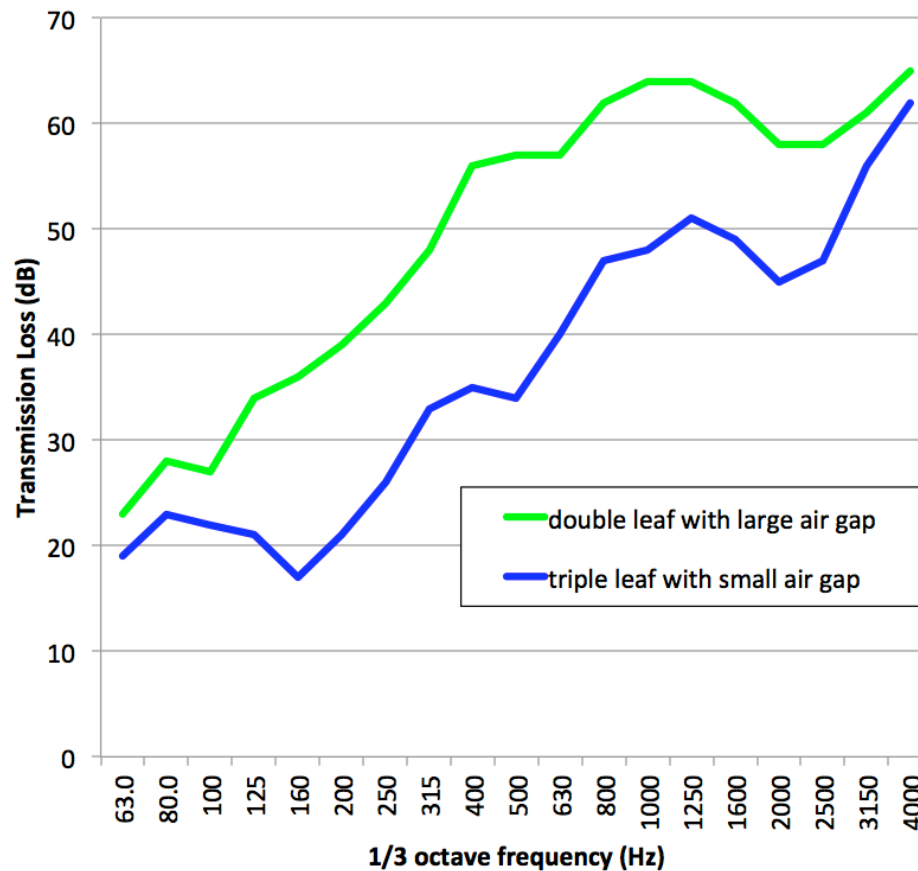
# TL OF THREE-LEAF PANELS

- *Three-leaf panels* have two air gaps
- Three-leaves (or more) are seldom used for noise control panels except when only high frequency performance is needed.
- While three-leaf panels provide an increase in high frequency TL, *the following potential problems develop*:
  - Low frequency TL usually decreases
  - Mass-air-mass resonance of small air gap reduces TL
  - It is difficult to determine the effect of multiple cavity resonances

# TL OF THREE-LEAF PANELS

Two vs. Three leaf partitions

- National Research Council Canada data

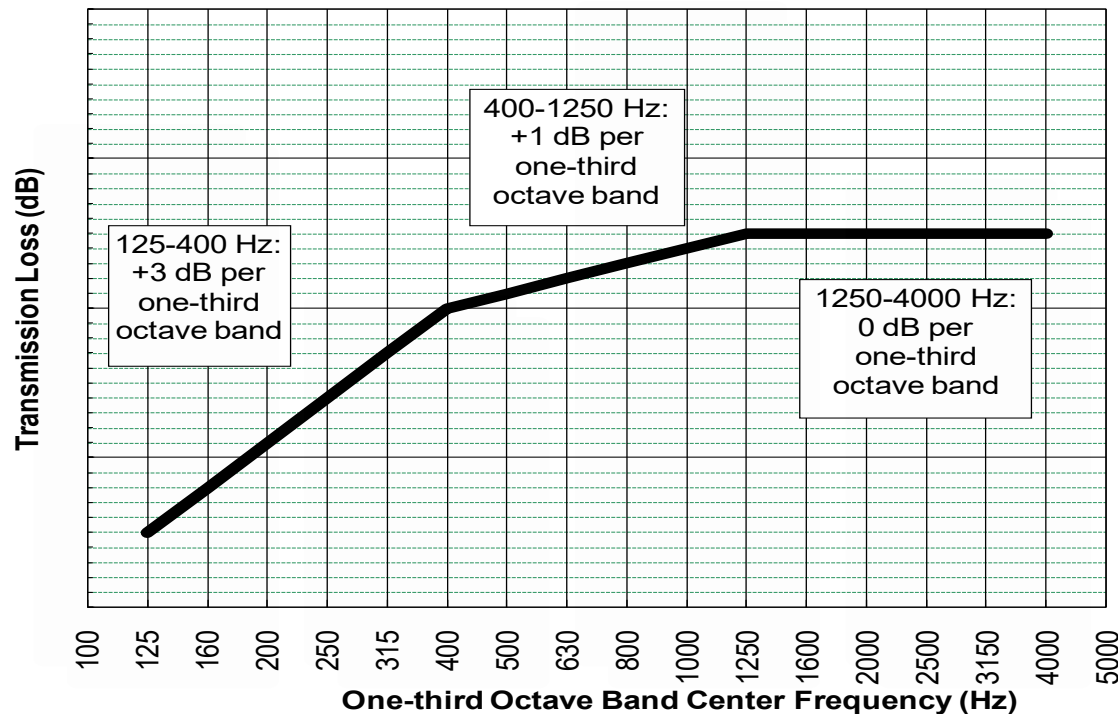


# AIRBORNE SOUND ISOLATION

SOUND  
TRANSMISSION  
CLASS (STC)

# SOUND TRANSMISSION CLASS (STC)

**STC is a single-number rating of the airborne sound transmission loss (TL) performance from 125 to 4000 Hz.** It is calculated based on a standard contour as the reference that is adjusted to the data per a standardized set of rules.



The higher the STC rating, the better the partition blocks sound

TL & STC data for common building elements is found on Egan p. 204-205.

# SOUND TRANSMISSION CLASS (STC)

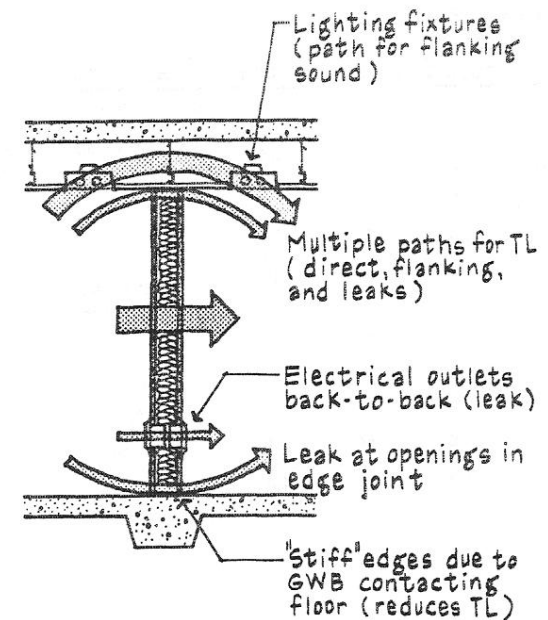
- The STC of the panel is the TL value at 500 Hz when a plot of the readings is overlaid on the standard plot under the following conditions:
  - Maximum deficiency (number of dB the data are below the standard contour) at any frequency shall not exceed 8 dB
  - Total sum of deficiencies of TL does not exceed 32 dB
- Adjust the STC contour to the highest position that satisfies the above two conditions. The STC rating is the numeric value where the contour crosses the 500 Hz ordinate.

# STC LIMITATIONS

- The frequency range of an STC test is from 125 to 4000 Hz. The low frequencies produced in music and by mechanical equipment are NOT represented in the test data.
- STC as **measured in a laboratory is under “perfect” conditions**. These are not seen in field conditions, and therefore could be misleading.

# LABORATORY vs FIELD CONDITIONS

- Laboratory STC test data (that you find in books) is measured under ideal conditions. There is **generally a degradation of 5-8 STC rating points** for construction in the field. Because...
  - Laboratory installations pass strict inspections and are expertly installed
  - No flanking paths or leaks exist where sound can get around or through the test specimen/partition
  - Poor workmanship in the field can cause sound leaks (the book claims up to 15 points of degradation is possible, though 5-8 dB is usually used to estimate effect)

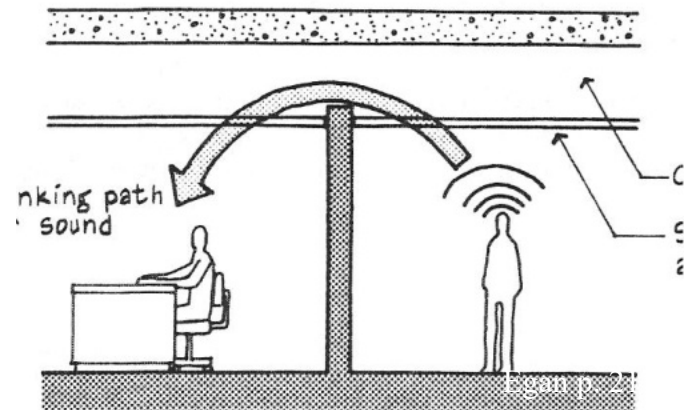


# STC VARIATIONS

- **ASTC – *Apparent Sound Transmission Class*** is a field measurement of STC and is lower because of flanking sound transmission.
- **FSTC – *Field Sound Transmission Class*** is older field measurement that is nearly identical to ASTC except for some testing technicalities.
- **OITC – *Outdoor-Indoor Transmission Class*** is determined exactly as STC, except over a wider range of frequencies at the 1/3 octave bands of 80 Hz to 4 kHz. The spectrum used is the average of transportation noises of vehicles, trains, and aircraft, which produce substantial low frequency sound. Used for building envelopes.

# STC VARIATIONS

- **CSTC – Ceiling Sound Transmission Class** (old)
- **CAC – Ceiling Attenuation Class** (currently in use)
  - Similar to STC except sound travels from a space through its ceiling, over the wall and through ceiling in the adjacent space. CAC is reported to the nearest 5 below the calculated CAC, e.g., a ceiling with a CAC of 38 would be classified as CAC 35 because it is between 35 and 39.
  - **Rule of thumb: High values of NRC correspond to low values of CAC** (recall that porous absorbers do not block sound). The exception would be when the ceiling tile is backed with a sound blocking layer.
  - For comparison, 1/2" gypsum board ceilings in adjacent spaces yield a CAC of 55.



# STC vs NOISE ISOLATION CLASS (NIC)

- STC is a normalized rating such that two partitions can be compared to each other
- The Noise Reduction (NR) per band can be used to calculate a single number rating called the **Noise Isolation Class (NIC)**.
  - *These values relate directly to the amount of sound reduction experienced by an occupant.*
  - Includes all flanking and leaks.
  - NR and NIC are different for each partition measured due to various flanking/leaks and the different conditions in the receiving room.
  - Can use it for “before” and “after” measurements.

# RELEVANT STANDARDS

- **STC (laboratory)** – ASTM E 90 *Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions*
- **ASTC/FSTC/NIC (field)** – ASTM E 336 *Standard Test Method for Measurement of Airborne Sound Attenuation between Rooms in Buildings*
- **CSTC/CAC (laboratory)** – ASTM E 1264 *Standard Classification for Acoustical Ceiling Products*
- **OITC (field)** - ASTM E1332 *Standard Classification for Determination of Outdoor-Indoor Transmission Class*

# Reference

- M. David Egan, Architectural Acoustics (Acton, MA: Copley Custom Textbooks, 2008) ISBN 1-58152-507-9
- Mechanical & Electrical Equipment for Buildings (MEEB), 12th Edition, Walter T. Grondzik and Alison G. Kwok, John Wiley and Sons, Inc., 2015, ISBN: 978-1-118-61690-4